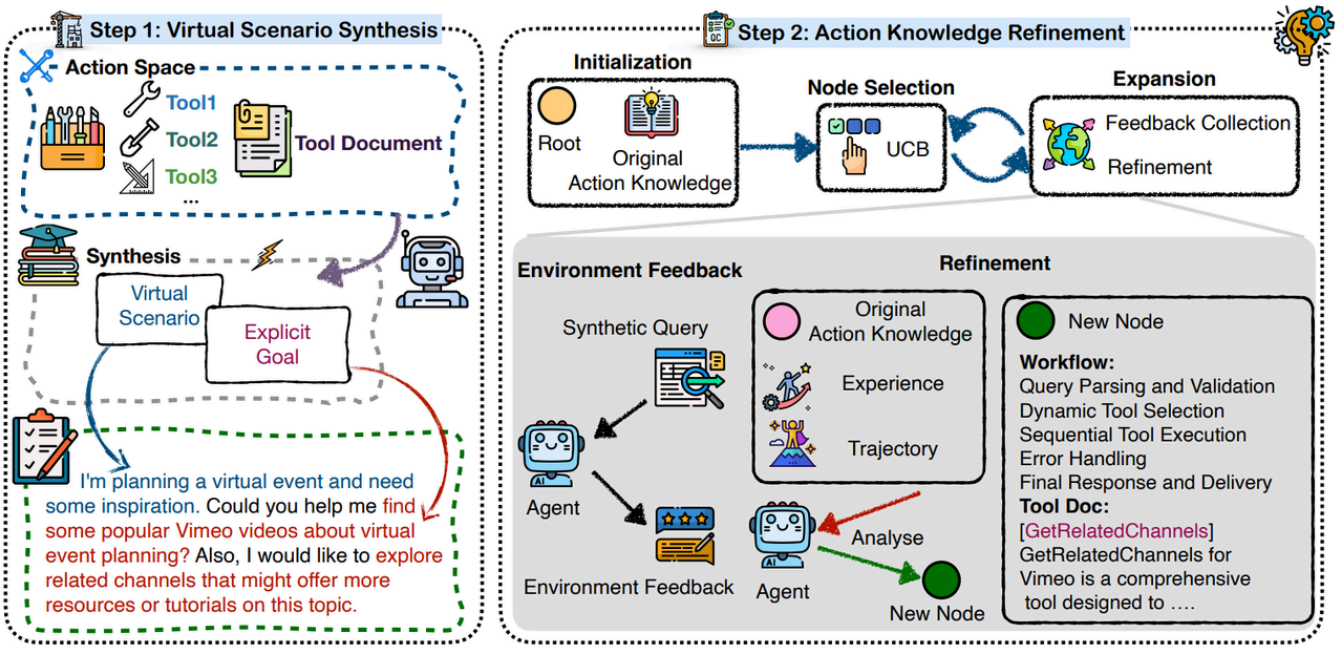
SynWorld: Virtual Scenario Synthesis for Agentic Action Knowledge Refinement

> #Tools #Dataset



- What problems does this framework solve?
  - When the LLM works in a completely new environment, its performance may not be stable (thus we need more environments).
  - Human-labeling is slow and expensive.
  - The current synthesized environments are mostly for single-step tool-use instead of complicated tool-use sequences.
- What does this framework do?
  1. Environment Preparation
     - With a tool set, sample a subset of tools.
     - With the subset of tools, let the LLM synthesize an environment with:
       - $\mathcal{B}$: The background (story) of the environment.
       - $\mathcal{G}$: The goal of the environment.
     - We want to ensure the diversity of the environment synthesized, so for each new environment, a similarity with the current environments will be calculated, and if the distance is less than a threshold, the newly generated one will be removed.
  2. Action Knowledge Exploration
     - With each environment, the agent will inference with MCTS used.
     - To select a new node, UCB is used to balance exploration and exploitation.
     - When a new node is selected, it will traverse back to query the results and the past optimization experience.
       - The experience, $\mathcal{E}$, contains scores before and after optimization, and the

modification made.
        - LLM will analyze the difference between the current situation and the historical experience, generating new knowledge about the action.
    - With experience, the model will finish the task and receive a new score. These will be used to update the experience and continue to guide the future exploration.
- Experiment Results
    - Experimental Setup: The authors conducted evaluations on two datasets: ToolBench (which examines multi-tool combination calls) and HotpotQA (which examines multi-step calls and planning for a single tool). Baseline comparisons included powerful methods such as ReAct and Self-Refine.
    - Key Finding 1: Overall superior performance. On ToolBench, SynWorld's pass rate and win rate both significantly outperformed other iterative optimization methods, demonstrating its superiority in optimizing tool combinations and task planning.
    - Key Finding 2: Strong knowledge generalization capability. Experiments show that the action knowledge learned by agents from synthetic data in virtual environments is highly scalable and can be effectively generalized to real-world applications.
    - Key Conclusion 3: Bidirectional optimization is indispensable. Ablation experiments reveal that the best results are achieved only when "Workflow" and "Tool Description" are optimized simultaneously (the two reinforce each other, providing high-quality negative examples or aligned environments).
- Work Cited
    - Paper
    - Chat History
    - NotebookLM